

Multiple Choice (10 marks)

1. A sample of solid MgCl_2 would be most soluble in which of the following solutions?
 - (a) $\text{AlCl}_3(\text{aq})$
 - (b) $\text{CBr}_4(\text{aq})$
 - (c) $\text{NaOH}(\text{aq})$
 - (d) $\text{HCl}(\text{aq})$
 - (e) $\text{LiOH}(\text{aq})$
2. 0.25 mol of a weak, monoprotic acid is dissolved in 0.25 L of distilled water. The pH was then measured as 4.26. What is the pK_a of this weak acid?
 - (a) 4.26
 - (b) 3.66
 - (c) 9.26
 - (d) 2.66
 - (e) 10.52
3. In the infrared spectrum of H^{127}I , there is an intense line at 2309 cm^{-1} . Calculate the force constant of H^{127}I .
 - (a) $400\text{ N} \cdot \text{m}^{-1}$
 - (b) $350\text{ N} \cdot \text{m}^{-1}$
 - (c) $250\text{ N} \cdot \text{m}^{-1}$
 - (d) $285\text{ N} \cdot \text{m}^{-1}$
 - (e) $450\text{ N} \cdot \text{m}^{-1}$
4. Calculate the reduced mass of a nitrogen molecule in which both nitrogen atoms have an atomic mass of 14.00.
 - (a) 1.75
 - (b) 3.50
 - (c) 14.00
 - (d) 10.00
 - (e) 12.00
5. Calculate the kinetic energy of an automobile weighing 2000 pounds and moving at a speed of 50 miles per hour. Record your answer in (a) Joules and (b) ergs.

- (a) 3.0×10^5 Joules, 3.0×10^{12} ergs
- (b) 2.5×10^5 Joules, 2.5×10^{12} ergs
- (c) 2.0×10^5 Joules, 2.0×10^{12} ergs
- (d) 1.8×10^5 Joules, 1.8×10^{12} ergs
- (e) 3.56×10^5 Joules, 3.56×10^{12} ergs

True / False (10 marks)

Answer True or False

6. True or False: The simplest alkene has a tetrahedral configuration around its double bond carbon atoms.
7. True or False: The number of allowed energy levels for the ^{55}Mn nuclide is 5.
8. True or False: The constants in the Dieterici equation are $a' = 2 RT_c V_c$ and $b' = V_c / 2$, derived from critical constants.
9. True or False: During the electrolysis of molten magnesium bromide, magnesium is produced at the cathode and bromine at the anode.
10. True or False: A light pulse of 2.00 mJ at a wavelength of 1.06 m contains approximately 1.80×10^{16} photons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the orbital angular momentum quantum number, l , in the context of ionization energy, particularly focusing on the electron that is most easily removed from ground-state aluminum.
12. Discuss how the decrease in atmospheric pressure at 4000 meters altitude affects the boiling point of water, and calculate the expected boiling point under these conditions.

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